Epic 3: Data Pre-processing

Description

This phase focuses on preparing the Credit Card Approval Prediction dataset for machine learning by improving its quality, consistency, and usability. Raw datasets often contain duplicate records, missing values, inconsistent formats, and categorical attributes that must be processed before model training. Effective preprocessing ensures that the data accurately represents applicant information and credit history, leading to better model performance and reliable prediction results.

During this phase, duplicate records are identified and removed, missing values are analyzed and handled appropriately, and data cleaning operations are performed to eliminate unnecessary or inconsistent information. Feature engineering techniques are applied to create meaningful attributes that improve prediction accuracy, while categorical variables such as education, occupation, housing type, and income type are converted into numerical representations suitable for machine learning algorithms.

The cleaned applicant dataset and credit history dataset are then merged into a single structured dataset that is ready for training and evaluation. Proper preprocessing reduces noise, minimizes bias, improves feature quality, and enables classification algorithms to learn meaningful patterns from the data.

Activities Performed

• Removing duplicate records from the dataset.

• Detecting and handling missing values.

• Cleaning and merging applicant and credit history data.

• Performing feature engineering to create meaningful variables.

• Encoding categorical features into numerical values.

• Preparing the final dataset for machine learning model training.

Outcome

After completing this phase, the dataset becomes clean, consistent, and machine-learning ready. High-quality preprocessing improves the performance, stability, and prediction accuracy of the Credit Card Approval Prediction system and provides a reliable foundation for model building.